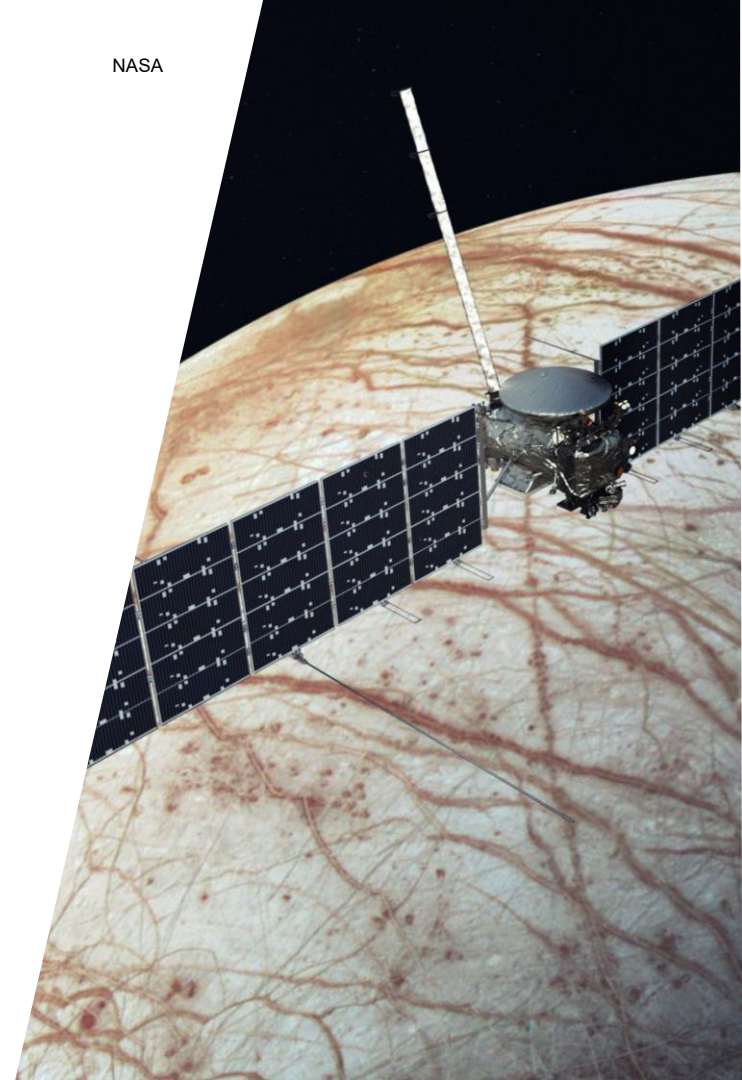


EUROPA CLIPPER

Space Systems Mission Study
eo23564



Mission aim

Aim ID	Mission Aim [1]
A1	Investigate ice shell and ocean nature.
A2	Analyse surface and atmosphere composition.
A3	Characterise geological features.

- The Europa Clipper is NASA's flagship mission to explore Jupiter's icy moon Europa, one of the most promising places in the solar system to search for conditions suitable for life.
- Although the Europa Clipper will not directly search for life, its objective is to assess whether Europa possesses the essential components: water, energy, and chemical building blocks, required to sustain life as currently understood.
- This mission is the first dedicated effort to conduct detailed flybys and evaluate Europa's ice thickness, ocean composition, and geological activity [1].

Stakeholder table

Stakeholder	Influence level	Role	Contributions
NASA HQ	High	Primary funder, mission owner, oversight of science goals.	Provides over \$5 billion in funding; Defines aims and expectations
NASA JPL	High	Project management, operations, integration, and testing.	Leads overall mission development, system-level testing. [6]
Johns Hopkins University APL	High	Spacecraft integration, testing, and key hardware provider.	Partners with JPL on mission development; Designs and integrates propulsion module, ECM, PIMS and radiation vault [10]
External Research Team	Medium to high	Pre-mission studies that inform refine mission design, perform lab experiments and simulations and modelling	Collaborative expertise from institutions like Arizona State University [26], University of Texas [27] Contribute to data interpretation, such as modelling ocean salinity from magnetic induction data and assessing biosignatures from surface composition maps.
Other Instrument Providers	medium	Specialized instrument development and integration for remote sensing and data collection.	E.g.: Teledyne’s detector for MISE infrared spectroscopy of chemical compositions [8], VIAVI Solutions’ precision filter for MISE to enable high-resolution mapping [9],
SpaceX	Medium to high	Launch services provider.	Provided Falcon Heavy rocket for launch on 14/10/2024, from Kennedy Space Centre [5]
Southwest Research Institute (SwRI)	medium	Instrument development and science leadership for composition and atmospheric studies.	Developed and delivered MASPEX, SUDA, and UVS; Supports data analysis for habitability indicators like water vapour and salts. [7]
Advanced Material Development	medium	UK-based nanomaterial technology provider for electromagnetic compatibility.	Supplying thin-film coatings for radar and instrument interfaces; E.g., Enhances compatibility for REASON by reducing electromagnetic interference. [5]
Public, Media & Planetary Society	low	Advocacy, outreach, and public engagement.	Supports grassroots campaigns and educational initiatives; Promotes public awareness of astrobiology potential, such as Europa’s subsurface ocean as a target for life detection, through media coverage and society-led petitions for mission funding.

Payload

Instrument Name [2]	Purpose	Instrument Type [2]	Aim	Mass (kg) [22]	Power (W) [22]	Resolution
EIS [29]	Characterize Europa's surface geology, landforms, and compositional variability; search for signs of recent or current activity, including plumes; constrain ice shell thickness and shape	Cameras	A3	66	253	100 m/pixel, 8MP
E-THEMIS [15]	Map Europa's thermal inertia, surface roughness, and block abundance; detect and characterize thermal anomalies linked to recent or ongoing geologic activity; constrain regolith properties and ice-shell thermal state	IR Imager	A3	16	44	1280*1280, 12 μ m
Europa-UVS [18]	Search for and characterize active plumes on Europa by detecting auroral emissions and atmospheric escape; map distribution and composition of surface frosts; determine composition and chemistry of atmosphere and ionosphere; and investigate ocean-atmosphere-surface exchange processes.	UV Spectrograph	A1	20	37	<0.6 nm (point source), <1.2 nm (extended source); resolving power $\lambda/\Delta\lambda = 220$; 0.16 degrees (Airglow Port), 0.06 degrees (High-spatial-resolution Port)
MISE [19]	Assess Europa's habitability by mapping distribution of surface compounds, including salts, organics, acids, radiolytic products, and trace materials; understand geologic history of the surface; and search for evidence of current geologic activity through compositional signatures.	IR Spectrometer	A1	64	63	25 m per pixel at 100 km altitude, and 300 cross-track pixels

Payload

Instrument Name	Purpose	Instrument Type	Aim	Mass (kg) [22]	Power (W) [22]	Resolution
ECM [30]	Characterize time-varying magnetic field induced by Europa's interior ocean in response to Jupiter's time-varying magnetic field; map magnetic field of the solar wind and Jupiter's magnetosphere; characterize magnetic field of Jupiter's extended neutral clouds; characterize magnetic field of the Europa plasma interaction.	Magnetometer	A1	34	23	24-bits and a 128 sample/second cadence
PIMS [25]	Measure plasma environment surrounding Europa to determine the composition, density, temperature, and flow of Jupiter's magnetospheric plasma; characterize rate and composition of material sputtered from Europa's surface; detect and characterize plumes by measuring their charged constituents and local plasma disturbances.	Plasma Instrument	A2	16	37	<5° for signals above 100 pA/cm ²
REASON [31]	Determine ice-shell thickness and large-scale structure; Map 3D distribution and character of water within the ice shell; Characterize ice shell composition and physical properties; Search for shallow subsurface water bodies; Investigate exchange processes between the surface, ice shell, and subsurface ocean;	Radar	A1	70	90	VHF: ≤30 m in ice (finer) in top 3 km, HF: ≤300 m in ice to 30 km depth, HF/VHF: ≤300 m in ice (coarser) from 3 km to 30 km depth
MASPEX [37]	Measure composition of Europa's exosphere, plumes, and atmosphere by detecting and identifying neutral gas species, including major volatiles, organic compounds, trace species, and potential biosignatures	Mass Spectrometer	A2	65	91	≥4275 (10% valley) up to 500 m/z
SUDA [20]	Sample and analyze composition of ice grains and dust ejected from Europa's surface through fractures or plumes; measure abundance and composition of salts, organics, and other ocean-derived materials in the ejecta; provide evidence of Europa's habitability by detecting potential biosignatures and geochemical tracers in freshly exposed material.	Dust Analyzer	A2	15	25	m/Δm ~ 200

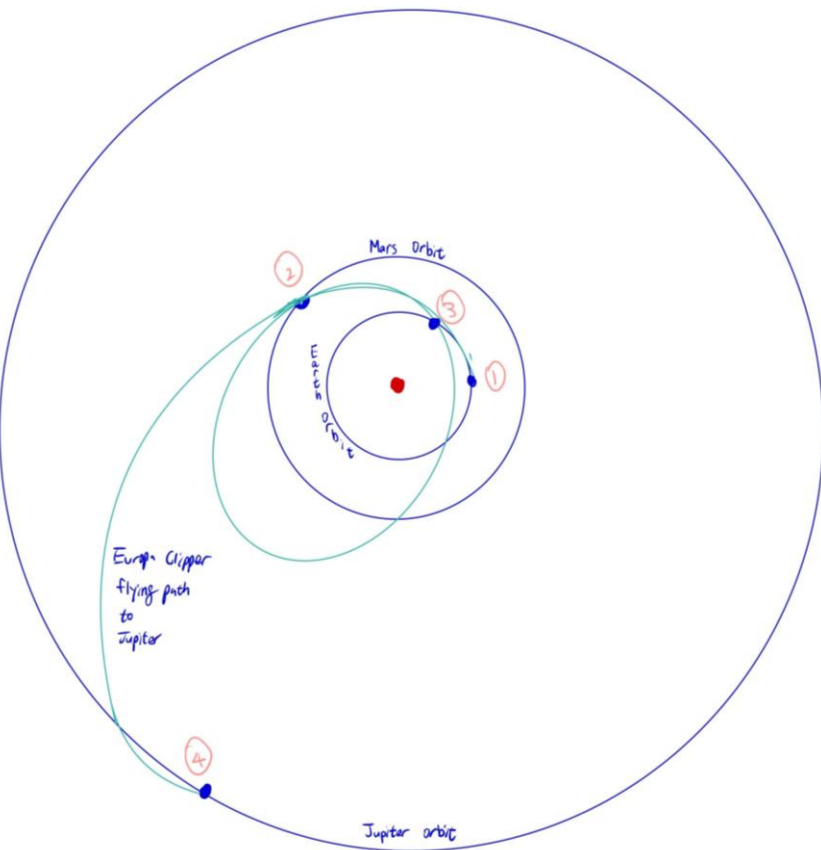
Requirements

ID	Requirement Text	Rationale Text [1]	Requirement Type	Traceability	Verification Method(s)
R1	The Europa Clipper spacecraft shall perform 49 flybys of Europa at an altitude of 50 to 2700km	Enable repeated data collection over mission duration to study temporal variations.	Mission	A1 + A2 + A3	Analysis and Test
R2	The Europa Clipper mission shall measure the thickness of Europa's ice shell with an accuracy of 1 km.	Confirm presence and characteristics of a subsurface ocean to assess habitability	Functional	A1	Test and Analysis
R3	The Europa Clipper spacecraft shall map at least 90% of Europa's surface composition.	Assess geology and chemical composition for insights into surface processes and potential biosignatures.	Functional	A2 + A3	Inspection and analysis
R4	The Europa Clipper spacecraft shall detect the presence of a subsurface ocean using magnetic induction measurements with a sensitivity to induced fields at Europa's synodic rotation period and orbital period, achieving a measurement precision of at least 1 nT.	Enable habitability assessment by confirming existence of a conductive subsurface layer indicative of a salty ocean.	Functional	A1	Analysis
R5	The Europa Clipper spacecraft shall withstand a total ionizing radiation dose greater than 2.9 Mrad over mission lifetime.	Ensure operational reliability in Jupiter's intense radiation environment.	Environmental	A1 + A2	Test and analysis

Requirements

ID	Requirement Text	Rationale Text	Requirement Type	Traceability	Verification method(s)
R6	The Europa Clipper spacecraft shall collect magnetic field data at a sampling rate greater than 1 Hz during flybys.	Determine ocean depth and salinity through high-resolution magnetic induction analysis [1]	Functional	A1	Inspection and test
R7	The Europa Clipper mission shall resolve tidal flexing of Europa's surface to an accuracy of 0.1 mGal.	Assess internal structure, including ice shell thickness and ocean properties.	Functional	A1 + A2	Analysis
R8	The Europa Clipper spacecraft shall synchronize all onboard instruments to within 1 second during flyby operations.	Ensure coordinated and time-aligned data collection for multi-instrument analysis during close approaches.	Operational	A1 + A3	Test and analysis
R9	The Europa Clipper spacecraft shall image Europa's geological features at a resolution of 0.5 m per pixel during selected flybys.	Capture detailed surface features to study geological processes and identify sites of potential recent activity.	Functional	A1 + A2	Inspection
R10	The Europa Clipper mission shall downlink at least 1.6 Tb of science data to Earth over the mission lifetime.	Transmit comprehensive datasets for detailed Earth-based analysis of Europa's habitability.	Operational	A1 + A2	Demonstration

CONOPS



Jupiter Arrival [3] [39]

Event 1: Launch

Accomplished Date: 14/10/2024

Activities: Took off from Launch Complex 39A on a Falcon Heavy rocket, entering Earth's orbit. The spacecraft separated shortly after, unfolded its solar panels, and ran basic equipment health checks.

Event 2: Mars Gravity Assist [4]

Accomplished Date: 1/3/2025

Activities: After several months of cruising through space, the spacecraft performed small engine burns to precisely adjust its trajectory, leading to a close flyby of Mars at about 884 km above the surface. This gravitational pull helped slow down, slinging it onward toward Earth while conducting some data collection on the Martian environment.

Event 3: Earth Gravity Assist

Estimated Accomplish Date: 3/12/2026

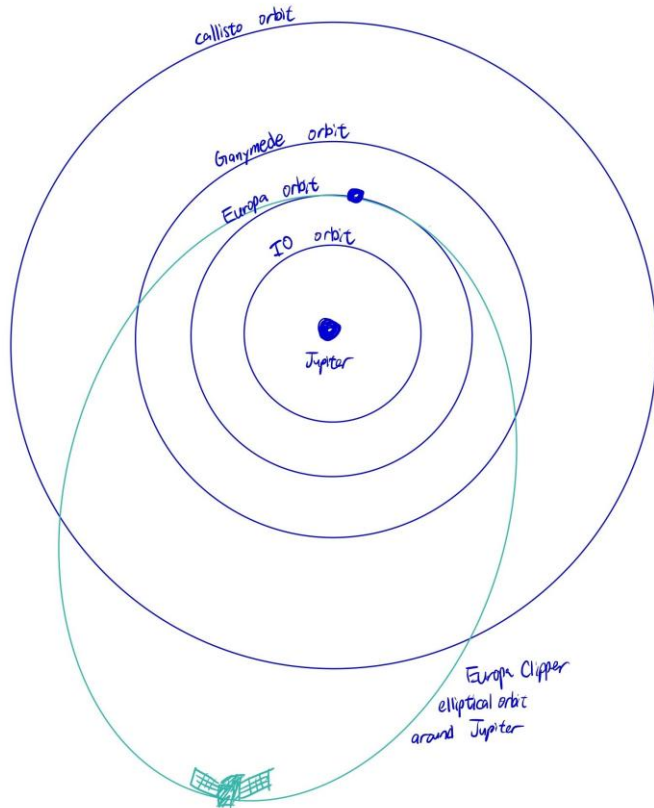
Activities: The spacecraft will swing by Earth at around 3200 km altitude, harnessing the planet's gravity to gain a significant speed boost. Along the way, it will execute further path corrections, conduct deep space navigation adjustments to refine its course for the long journey ahead.

Event 4: Jupiter Orbit Insertion

Estimated Accomplish Date: 11/4/2030

Activities: The spacecraft will use a flyby of Jupiter's largest moon, Ganymede, to assist in entering orbit around the gas giant. Engines will fire to establish a wide, elliptical path designed to minimize time in Jupiter's intense radiation belts, followed by efforts to stabilize the orbit and start preliminary surveys of the Jovian moons. The process should take 6 hours.

CONOPS



Science Operations [3]

Activities: In orbit around Jupiter, the spacecraft will make 49 close flybys of Europa, coming as near as 25 kilometers from its surface. These passes, starting 25/5/2031 and occurring every two to three weeks, will use onboard tools to study the ice layer's thickness, the moon's magnetic field, surface chemicals, and gravity clues about the hidden ocean.

Estimate accomplish date: Completion of 49th flyby on August 10, 2034.

Duration: 3.3 years.

Off-Nominal: Instrument anomaly during flyby (e.g. latchups)

Contingency: Switch to redundant sensors; autonomous safing stops data collection and reorients for comms. If close approach is too risky, abort flyby, use engines to evade, and reschedule through ground update.

Data Downlink [32]

Activities: spacecraft sends back data to Earth via NASA Deep Space Network's big antennas in California, Australia, and Spain. Over time, it will transmit images and readings for data analysis.

Off-Nominal: Transmission failure (e.g. antenna misalignment)

Contingency: Autonomous re-pointing using star trackers; retry via backup low-gain antenna. Ground commands for re-transmission of buffered packets.

Decommissioning [3]

Activities: after science wraps up, the probe will be steered into a crash on Ganymede. This ensures it's destroyed and doesn't risk contaminating Europa. A final check-in with Earth confirms the end.

Estimate accomplish date: 3/9/2034

Potential Extended Mission Scenarios

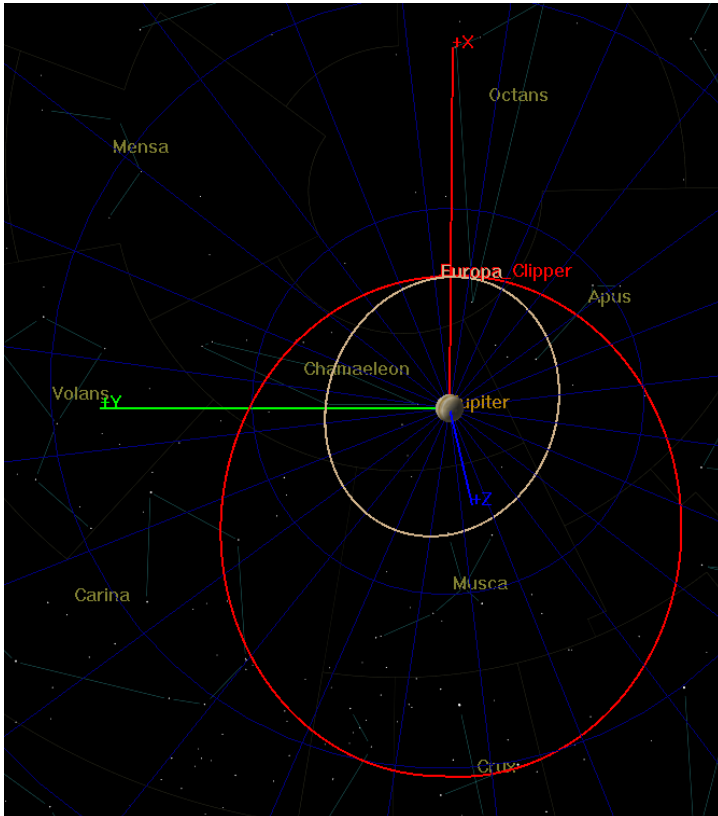
Scenario 1: Additional Flybys: If fuel permits, extend to 60 flybys.

Scenario 2: Ganymede Study: Shift focus to Ganymede orbits post-Europa.

Scenario 3: Callisto Encounter: One-time flyby if trajectory aligns.

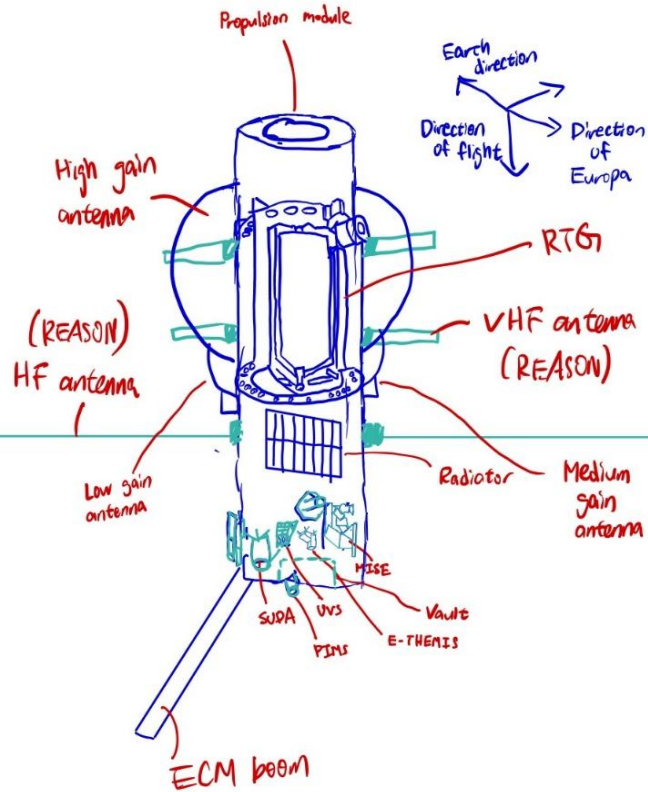
Orbit model (GMAT)

Element	Value
SMA	1354993
ECC	0.5074
INC	0.2095
RAAN	0
AOP	359.1563
TA	359.6517



- The selected orbital elements model a Europa flyby trajectory that satisfies Requirement R1.
- A semi-major axis of 1,354,993 km and an eccentricity of 0.5074 result in a periapsis near 756,000 km, just outside Europa's orbit. This enables low-altitude flybys with minor timing adjustments, while an apoapsis of approximately 2.31 million km reduces radiation exposure from Jupiter.
- An inclination of 0.2095° closely aligns with Europa's orbit, allowing efficient repeated targeting.
- A right ascension of the ascending node of 0° and an argument of periapsis of 359.1563° establish consistent inertial orientation.
- Setting the true anomaly to 359° places the spacecraft nearly at periapsis at the start of the animation, corresponding to the moment of closest approach during a Europa flyby. The previous flyby would have occurred one orbit earlier, demonstrating that this orbit supports multiple sequential flybys.

Alt. Config. 1



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Design:

- No solar panels
- Use radioisotope thermoelectric generator (RTG), like the Cassini project [33]

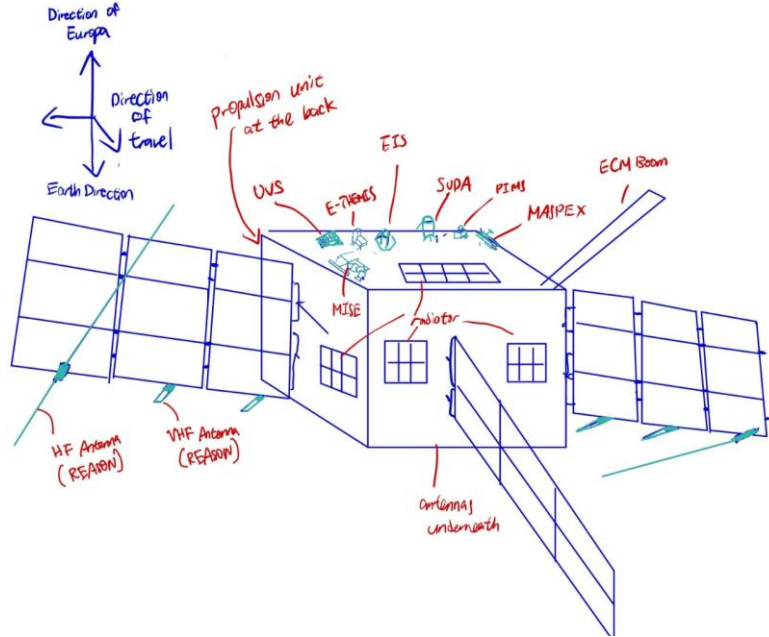
Drivers of using this design:

- RTGs provide constant, **steady power day and night** without needing batteries for storage. For a mission like Europa Clipper, which has instruments running continuously and needs thermal control, that reliability is crucial.
- RTG is **radiation-resistant** and can survive the harsh Jupiter radiation environment for the whole mission.
- RTGs **produce waste heat**, which can help keep instruments warm in the outer solar system. Solar arrays don't help with this, so extra heaters would be needed.
- Unlike solar arrays, RTGs have **no moving parts** that need to deploy. This eliminates the risk of a mechanism failing or sticking, which could jeopardize the mission.

Rationale for Excluding the RTG Design:

- Because of advanced solar technology, current solar panel designs can **sufficiently power** the spacecraft even at Jupiter's distance from the Sun.
- Using RTG would have been significantly more **expensive**, potentially adding tens of millions of dollars to the project budget, while solar arrays provide comparable performance at a lower cost and with similar mass implications.
- **Solar panels degrade more slowly** than RTG in this mission profile, enabling potential mission extensions, and they align with the spacecraft's orbit design, which keeps it mostly in sunlight.
- NASA may have faced **plutonium shortages** for RTGs at the time of planning, further favouring solar as a more readily available option

Alt. Config. 2



Design:

- Satellite bus a cube instead of cylinder
- Essentially a **giant CubeSat**

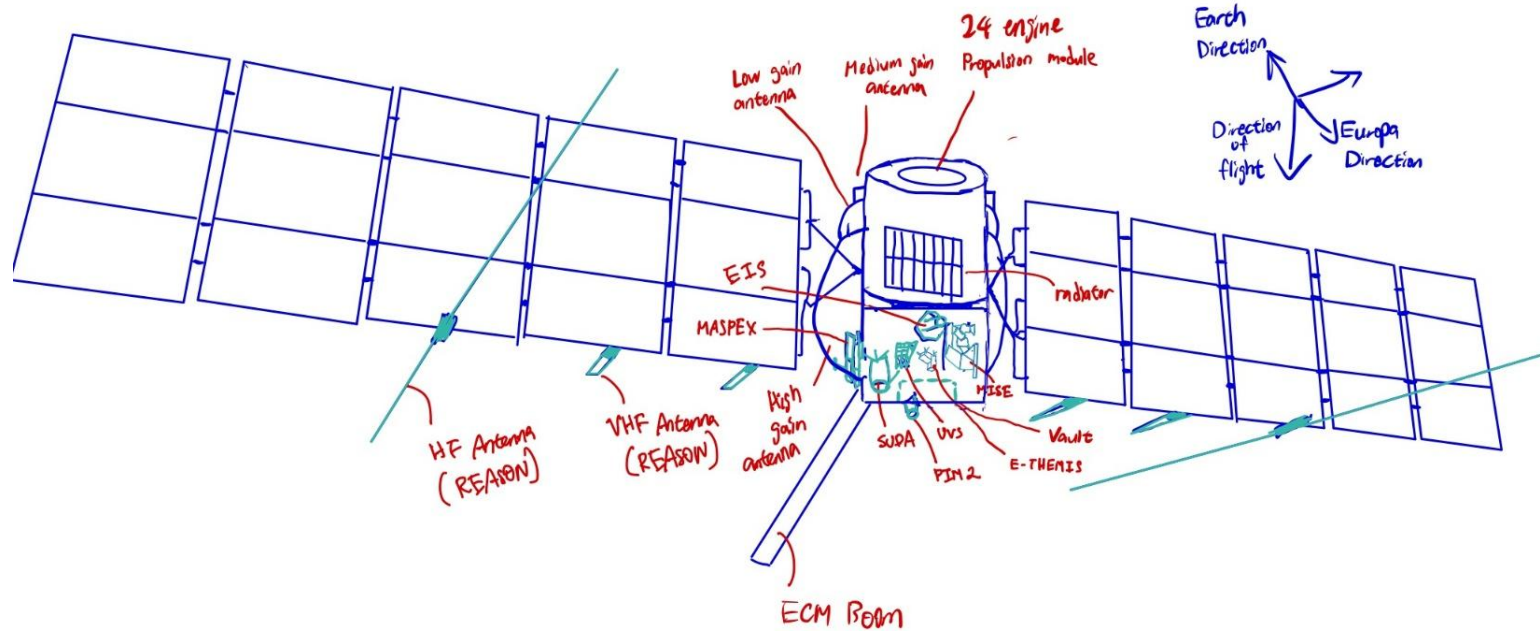
Design Drivers:

- **Thermal control:** The large CubeSat shape lets heaters and insulation be placed evenly, which is important when operating far from the Sun or near Jupiter's radiation belts.
- **Radiation protection:** The cube shape makes it easier to evenly shield sensitive electronics and instruments.
- **Instrument integration:** The cubic design lets multiple instruments and sensors be arranged efficiently around a central bus, which makes wiring and structural support simpler.
- **Attitude control:** A cube has balanced axes, so its spin and pointing are more predictable and use less fuel than a long cylinder.

Rationale for Excluding the CubeSat Design:

- **Limited Deep-Space Heritage:** CubeSats historically high-performance propulsion, and reliable deep-space communication developments, giving them very little long-duration flight heritage; this may make NASA reluctant to rely on them for critical flagship-level science missions.
- **Structural Reinforcements:** A CubeSat sent to Jupiter would need substantial structural reinforcement to survive launch, long duration deep space stresses, and Jupiter's environment, which increases total mass, raising both launch cost and required fuel.
- **Fuel Tank Geometry:** A typical cylindrical, domed-end fuel tank like in the real configuration would waste internal volume in a compact CubeSat style spacecraft; designing a highly customized, space-efficient tank shape to fix this would significantly increase manufacturing complexity and overall mission cost.

Chosen Configuration



- 2 giant solar array for simple deployment and cost saving considerations
- Cylindric bus for simplicity and reliability, concluded from previous deep space missions
- 24 small thrusters for propulsion to satisfy high delta-v needs

Thermal design

- The satellite was assumed to be a spherical, grey bodied (Stefan law) and coated in polished aluminium, with radius= 2m
- Surface area = 50.27 m², projection area = 12.56 m²
- Polished Aluminium solar absorbance $\alpha = 0.2$, IR emissivity $\varepsilon = 0.05$ [28]
- Assume radiator perpendicular to solar vector
- At hot case, assumed earth catapult where satellite altitude = 3200 km [3]

Parameter	Jupiter (Cold case)	Earth (Hot case)
Distance from Sun d	5.2 AU [17]	1 AU
Solar flux G _s	50.5 [16]	1370 [28]
Planet radius	71492 [16]	6371
View factor	6.29 e-4	0.1267
Internal Power dissipated	10 W [21]	350 W [21]
albedo	0.52 [16]	0.3

Table for environmental parameters

Table of satellite instrument temperature limits

Component	Operating Limit (°C)	Survival Limit (°C)
EnerSys® ABSL Battery [14]	0 to 40, [13]	-20 to 60, [13]
Electronics (E-THEMIS)	-7 to 35, [15]	-20 to 50, [15]
MMH/NTO Propellant [12]	15 to 27	-52 (MMH) & -10 (NTO)

Cold case calculation:

$$q_{ir} = \varepsilon * \sigma * A_s * T^4 = 0.05 * 5.67 * 10^{-8} * T^4 = 2.835 * 10^{-9} T^4$$

Heat loss = internal power dissipation + Jupiter IR radiation

$$\varepsilon * \sigma * A_s * T^4 = P + q_{ir} * \varepsilon * A_s * F_{psc}$$

$$0.05 * 5.67 * 10^{-8} * 50.27 * T^4 = 10 + 2.835 * 10^{-9} * 0.05 * 50.27 * 6.29 * 10^{-4}$$

$$9.255 * 10^{-8} T^4 = 10 + 4.48 * 10^{-12} T^4$$

$$9.255 * 10^{-8} T^4 = 10$$

$$T^4 = 1.08 * 10^6$$

$$T = 101 \text{ K} = -172 \text{ °C}$$

Thermal design

Evaluation:

- **Hot-case equilibrium temperature:** 145 °C **significantly exceeds** spacecraft survival limits. It would destroy the spacecraft during second earth catapult, risking component failure.
- **Cold-case equilibrium temperature:** -172 °C is **far colder** than the survival limits in Jupiter eclipse, causing propellant freeze, battery & electronics shutdown, and potential structural brittleness.
- **Radiator area:** 3.46m² is **acceptable** for a 3*4*5 satellite, however significantly larger than actual value of 1.29 m²

Adopted hybrid passive/active architecture in Europa Clipper [21]:

- **Heat Rejection System (HRS):** A thermal pump module with two mixing valves for thermal control. The design turns down heat loss by 35:1.
- The two **Thermal Control Valves** were integrated in series for radiator bypass, reducing flow in cold case.
- **Radiator emissivity changes** by louver from 0.14-0.74
- Supplement with **MLI blankets** and survival heaters for eclipse.
- Implementing these yields operational temperatures 0 °C to 30 °C vault-wide, meeting temperature requirements.

Hot case calculation:

$$q_{ir} = \varepsilon \sigma A_s T^4 = 0.05 * 5.67 * 10^{-8} * T^4 = 2.835 * 10^{-9} T^4$$

Heat loss = internal power dissipation + direct solar input + earth albedo + Jupiter IR radiation

$$\varepsilon \sigma A_s T^4 = P + \alpha A_{proj} G_s + \alpha A_s G_s \alpha F_{p-sc} + q_{ir} \varepsilon A_s F_{p-sc}$$

$$0.05 * 5.67 * 10^{-8} * 50.27 * T^4 \\ = 350 + 0.2 * 12.56 * 1370 + 0.2 * 50.27 * 0.3 * 0.1269 * 1370 + 2.835 * 10^{-9} * 0.05 * 50.27 * 0.1269 * T^4$$

$$9.255 * 10^{-8} T^4 = 350 + 3441.44 + 524 + 9.04 * 10^{-10} T^4 \\ 1.416 * 10^{-7} T^4 = 4315.44 \\ T^4 = 3.048 * 10^{10} \\ T = 417.8 \text{ K} = \mathbf{144.8 \text{ } ^\circ\text{C}}$$

Radiator area:

- Assume radiator perpendicular to solar vector
- Radiator emissivity of louvre going up to 0.74 in hot case [21], and absorptivity down to 0.1 (assumption)
- Uses hot case since it is where radiator performance is most critical
- Since radiator is a plate, $A_s = A_{proj}$
- Assume Q_w = total heat loss of hot case

$$\varepsilon \sigma A_s T^4 = Q_w + \alpha A_{proj} G_s \\ A_s * 0.74 * 5.67 * 10^{-8} * 416^4 = 4342 + 0.1 * 1370 * A_{proj} * \cos 90 \\ 1256.6 A_s = 4342 \\ A_s = \mathbf{3.46 \text{ m}^2}$$

OBDH design

All avionics systems are equipped with two sets of assemblies to provide redundancy in the event of a fault. The backup unit remains powered off, except during scheduled activations to back up critical scientific data and record the spacecraft state. [22]



[22]

OBDH design

Data Communication Methods:

- **Space Wire:** High-speed serial links, low power consumption, deterministic low latency. Chosen for most science instruments, so that data can flow quickly to the BDS without bottlenecks, plus the Telecommunications subsystem's high-volume telemetry with avionics[38].
- **RS-422:** Excellent noise immunity, high common-mode rejection, reliable in radiation-heavy environments. Selected for subsystems and sensors that are physically far from the vault or need magnetically clean connections: Thermal Control Unit, Propulsion Module, and Radiator Monitor Sensor Assembly. [36]
- **UART:** Minimum hardware complexity, extremely low power, zero routing overhead. Used for very low-rate instruments, such as PIMS. [34]
- **MIL-STD-1553B bus:** Extremely robust, time-tested, dual-redundant command/response bus with strong error detection. Used for critical spacecraft housekeeping and fault-tolerant commanding of major subsystems (power distribution, thermal control unit, propulsion module), where redundancy and strict command verification are more important than speed. [35]

Why Only X-Band and Ka-Band for Communications? [23]

- **X-Band for Uplink/Downlink Reliability:** Chosen for bidirectional commands and low-rate telemetry due to mature NASA Deep Space Network support, atmospheric penetration, and low susceptibility to Jupiter's plasma interference; enables safe, real-time ops over 45-min light travel time.
- **Ka-Band for High-Rate Science Downlink:** Higher frequency yields greater bandwidth efficiency at 5 AU distances, but not used simultaneously with X-band to avoid hardware conflicts.
- **Minimisation Rationale:** Dual-band setup balances reliability X with throughput Ka without lower bands that suffer higher attenuation in Jovian environment, reduces antenna needs

Security Need	Implementation on Europa Clipper
Protection of safety-critical data	Aluminium-zinc vault enclosing all electronics, preventing any external interference or damage.
Protection of mission data from unauthorised access	All data sent back from the spacecraft will be encrypted to prevent unauthorized access. NASA has implemented robust security protocols to protect the data throughout the entire mission lifecycle. [24]
Protect operationally sensitive payload data	Ground station systems to verify the integrity of the data, ensuring that it has not been altered or corrupted during transmission.

Verification table

ID	Requirement	Verification Method(s)	Verification Approach
R1	The Europa Clipper spacecraft shall perform 49 flybys of Europa at an altitude of 50 to 2700km	Analysis Test	What: Verify the spacecraft performs 49 flybys at specified altitudes How: Full-mission trajectory propagation analysis in GMAT (or other simulation software); Test GNC subsystem in hardware-in-the-loop simulator to confirm altitude control during flyby When: Subsystem/System levels
R2	The Europa Clipper mission shall measure the thickness of Europa's ice shell with an accuracy of 1 km.	Test Analysis	What: Verify if satellite can measure ice shell thickness with 1 km accuracy How: Test REASON radar sounder against known ice layers in large ice-tank facility; Analyse end-to-end data processing chain using flight software on ground testbed When: Subsystem level
R5	The Europa Clipper spacecraft shall withstand a total ionizing radiation dose greater than 2.9 Mrad over mission lifetime.	Test Analysis	What: Verify if satellite can withstand total ionizing radiation dose greater than 2.9 Mrad over mission lifetime How: Test radiation vault and flight-spare avionics under radioactive material or proton irradiation to similar levels; Analyse radiation transport modelling results against measured test data When: Subsystem level

Validation table

ID	Aim	Type	Validation Method(s)	Validation Approach
A1	Investigate ice shell and ocean nature	Goal	Inspection	What: Confirm definitive detection and characterisation of a global subsurface ocean through mission data How: Inspection of peer-reviewed science publications arising from Europa Clipper data When: Post-mission
A2	Analyse surface and atmosphere composition	Goal	Inspection	What: Confirm mission science objectives for surface and exosphere composition are fully met How: Inspection of published global composition maps and atmospheric/exospheric models derived from mission datasets When: Post-mission
A3	Characterise geological features	Goal	Demonstration	What: Confirm high-resolution characterisation of geological features and evidence of recent activity How: Publishing high-resolution stereo topography, identify recent geological activity and plumes in peer-reviewed results When: Post-mission

Reflection

- The primary design drivers were requirements R1 and R5. Requirement R1 means large deployable solar arrays were required, as the spacecraft operates primarily beyond 5 AU, where solar flux is only 50.5 W/m^2 . It also required a high-delta-V propulsion system to repeatedly raise the periapsis above the intense radiation belts during flybys. Requirement R5 imposed even greater constraints by forcing creation of a 200+ kg aluminium-zinc radiation vault, which determined instrument placement, as all avionics must be within it, and significantly restricted configuration options.
- A simplified thermal analysis assumed a passive, grey-body spacecraft with an absorptivity of 0.2 and an emissivity of 0.05. This approach resulted in equilibrium temperatures of -172°C in the cold scenario and 145°C during the Earth gravity assist hot scenario, both of which exceed the survival limits for batteries and propellant. In contrast, published data on the Europa Clipper indicate that the vault is maintained at $0\text{--}30^\circ\text{C}$. Moreover, the radiator area was calculated to be 3.46m^2 , significantly larger than real value 1.29 . This is due to real design using a single-phase, mechanically pumped fluid loop and variable-emissivity louvers, as well as thermal control valves. The discrepancy arises from only using radiator to maintain temperature in the simplified model. Including the features in the real scenario would significantly reduce the required radiator area in the hot case and required power dissipation in cold power.
- Through research and review of NASA's instrument specifications and system architecture papers, I gained a deeper understanding of the Europa Clipper mission and artificial satellites. This technical knowledge will guide my future projects, especially in recognizing how real-world constraints, such as Jupiter's radiation environment, affect subsystem selection. I also improved my ability to distil complex information for PowerPoint presentations. Since I am more used to writing essays and reports, I initially found it challenging to present content concisely. This project highlighted the importance of clarity in reports and slide decks, particularly when summarizing complex topics such as OBDH design justification into bullet points without ignoring critical content. Additionally, I gained insight into systems engineering and the importance of effective coordination among subsystems. Overall, this hands-on experience shifted my perspective from focusing on individual subsystems to adopting a broader design perspective, emphasizing the importance of traceability and trade studies in developing feasible missions.

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- **[2]** NASA (n.d.) 'Europa Clipper instruments'. Available at: <https://science.nasa.gov/mission/europa-clipper/spacecraft-instruments/> (Accessed: 29 November 2025).
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